

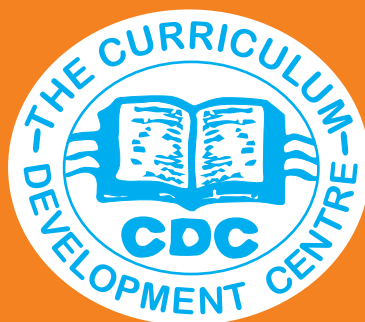


Republic of Zambia

MINISTRY OF EDUCATION, SCIENCE, VOCATIONAL TRAINING AND EARLY EDUCATION

CHEMISTRY SYLLABUS

GRADES 10 – 12



**PREPARED AND WRITTEN BY THE CURRICULUM DEVELOPMENT CENTRE
P.O. BOX 50092, LUSAKA – ZAMBIA
2013**

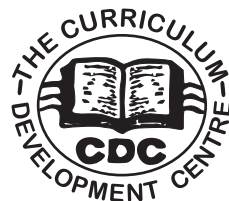


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VISION

Quality, lifelong education for all which is accessible, inclusive and relevant to individual, national and global needs and value systems

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PREFACE

The syllabus was produced as a result of the Curriculum review process carried out by the Ministry of Education, Science, Vocational Training and Early Education under the auspices of the Curriculum Development Centre (CDC). The curriculum reform process started way back in 1999 when the Ministry of Education commissioned five (5) curriculum studies which were conducted by the University of Zambia. These studies were followed by a review of the lower and middle basic and primary teacher education curriculum. In 2005 the upper basic education National survey was conducted and information from learners, parents, teachers, school managers, educational administrators, tertiary institutions traditional leader's civic leaders and various stakeholders in education was collected to help design a relevant curriculum.

The recommendations provided by various stakeholders during the Upper Basic Education National survey of 2005 and National symposium on curriculum held in June 2009 guided the review process.

The review was necessitated by the need to provide an education system that would not only incorporate latest social, economic, technological and political developments but also equip learners with vital knowledge, skills and values that are necessary to contribute to the attainment of Vision 2030.

The syllabus has been reviewed in line with the Outcome Based Education principles which seek to link education to real life experiences that give learners skills to access, criticize, analyse and practically apply knowledge that help them gain life skills. Its competences and general outcomes are the expected outcomes to be attained by the learners through the acquisition of knowledge, skills, techniques and values which are very important for the total development of the individual and the nation as a whole.

Effective implementation of Outcome Based Education requires that the following principles be observed: clarity of focus, Reflective designing, setting high expectations for all learners and appropriate opportunities.

It is my sincere hope that this Outcome Based syllabus will greatly improve the quality of education provided at Grade 8 and 9 as defined and recommended in various policy documents including Educating Our Future` 1996 and the `Zambia Education Curriculum Framework ` 2013.



Chishimba Nkocha
Permanent Secretary

MINISTRY OF EDUCATION, SCIENCE, VOCATIONAL TRAINING AND EARLY EDUCATION

ACKNOWLEDGEMENTS

The syllabus presented here is a result of broad-based consultation involving several stakeholders within and outside the education system.

Many individuals, institutions and organizations were consulted to gather their views on the existing syllabus and to accord them an opportunity to make suggestions for the new syllabus. The Ministry of Education wishes to express heartfelt gratitude to all those who participated for their valuable contributions, which resulted in the development of this syllabus.

The Curriculum Development Centre worked closely with other sister departments and institutions to create this document. We sincerely thank the Directorate of Teacher Education and Specialized Services, the Directorate of Planning and Information, the Directorate of Human Resource and Administration, the Directorate of Open and Distance Education, the Examinations Council of Zambia, the University of Zambia, schools and other institutions too numerous to mention, for their steadfast support.

We pay special tribute to co-operating partners especially JICA in collaboration with Hiroshima University and UNICEF for rendering financial and technical support in the production of this syllabus.



C.N.M Sakala (Mrs.)

Director-Standard and Curriculum

MINISTRY OF EDUCATION, SCIENCE, VOCATIONAL TRAINING AND EARLY EDUCATION

INTRODUCTION

This syllabus is designed for Grades 10 - 12. It is intended for learners taking Chemistry at Senior Secondary School Level of education. It places less emphasis on factual material and greater emphasis on understanding and application of scientific concepts and principles. This has been done so that learners develop skills that will be of the value for a long time in an increasingly technological world and it is expected that these will be of relevance for a very long time.

GENERAL AIMS

These provide the educational purposes of following a Chemistry Course at this level of education and are listed in a suggested order of priority:

The General aims are to:

1. provide, through well designed studies of experimental and practical Chemistry, a worthwhile educational experience for all learners, whether or not they go on to study Chemistry beyond this level and, in particular, to enable them to acquire sufficient understanding and knowledge to:
 - 1.1 become confident citizens in a technological world, able to take or develop an informed interest in matters of scientific importance
 - 1.2 recognise the usefulness, and limitations, of scientific method and to appreciate its applicability in other disciplines and in everyday life;
 - 1.3 be suitably prepared for studies beyond Senior Secondary School level in Chemistry, in applied Sciences or in Science dependent vocational courses.
2. stimulate learners, create and sustain their interest in the learning of Chemistry.
3. develop abilities and skills that:
 - 3.1 are relevant to the learning and practice of chemistry;
 - 3.2 are useful in everyday life;
 - 3.3 encourage efficient and safe practice;
 - 3.4 encourage effective communication.

4. develop attitudes relevant to Chemistry such as:
 - 4.1 concern for accuracy and precision;
 - 4.2 objectivity;
 - 4.3 integrity.
 - 4.4 appreciation
5. assist the development of the skills of:
 - 5.1 enquiry;
 - 5.2 initiative;
 - 5.3 inventiveness.
 - 5.4 Demonstration
 - 5.5 Analysis
 - 5.6 Stimulate interest in and care for the local and global environment.
6. promote an awareness that:
 - 6.1 scientific theories and methods have developed, and continue to do so as a result of co-operative activities of groups and individuals;
 - 6.2 the study and practice of Chemistry is subject to social economic, technological, ethnical and cultural influences and limitations;
 - 6.3 the applications of Chemistry may be both beneficial and detrimental to the individual, the community and the environment;
 - 6.4 Chemistry transcends national boundaries and that language of science correctly and rigorously applied, is universal.

MATHEMATICAL REQUIREMENTS

This syllabus offers a context in which mathematical skills and techniques may be applied in a relevant and more meaningful way. The study of Chemistry through this syllabus therefore strengthens the applications of Mathematics.

Candidates will be required to be competent in the following mathematical techniques:

1. add, subtract, multiply and divide.
2. use averages, fractions, percentages, ratios and reciprocals.
3. recognise and use standard notation.
4. use direct and inverse proportion.
5. use positive and negative indices.
6. plot graphs from given data.
7. interpret charts and graphs.
8. select suitable scales and axes for graphs.
9. make approximate evaluations of numerical expressions.
10. recognise and use the relationship between length, surface area and volume and their units on metric scales.
11. solve equations of the form $x = yz$ for only one variable when the other two are known.
12. make accurate numerical work and handle calculations up to three (4) significant figures.
13. comprehend and use symbols notation such as $\geq$ and $\leq$.

ASSESSMENT

Continuous assessment will be emphasised by using various methods of testing according to topics and themes at various levels. The examinations council of Zambia will prepare detailed procedures on how continuous assessment will be conducted by the teachers. The examination council will also develop examination syllabus to provide teachers with guidelines on the objectives to be tested. The scheme of assessment will consist of school based assessment and final examination that will be conducted by the examinations council of Zambia.

School based assessment will be in the form of tests. Tests will be in the form of diagnostic, aptitude, achievement, oral, practice attitude and performance, learners.

Assessment objectives

The following aspects of the aims will be assessed:

1. Knowledge with understanding

The candidates should be able to demonstrate knowledge and understanding in relation to:-

- (a) Scientific phenomena, facts, concepts, theories and laws.
- (b) Scientific terminology, use of symbols, quantities and units.
- (c) Scientific apparatus and instruments and their safe operations.
- (d) Scientific quantities and their determination.
- (e) Scientific and technological applications with social, economic and environmental relevance.

Questions testing these outcomes will in most cases begin with the terms such as:

“describe, discuss, state, explain, name, outline or define”.

2. Handling information and solving problems.

The candidate should be able to:-

- (a) Locate, select, organise and present information from a variety of sources.
- (b) Translate information from one form to another.
- (c) Manipulate numerical data.
- (d) Identify patterns and draw inferences from information.
- (e) Give reasonable explanations for patterns and relationships.

- (f) Make predictions and hypotheses.
- (g) solve problems

Questions testing these outcomes will often begin with the term such as “predict, calculate, or determine”.

3. Experimental Skills and Investigating.

The candidate should be able to:-

- (i) Follow instructions.
- (ii) Use basic laboratory techniques, apparatus and materials.
- (iii) Observe, measure and record.
- (iv) Plan investigations.
- (v) Interpret and evaluate observations and results.
- (vi) Predicts trends.
- (vii) Evaluate methods and suggest possible improvements.

TIME ALLOCATION

A minimum of six teaching periods of forty (40) minutes each per week. Preferably two (2) double periods to be taken in the Laboratory.

GRADE 10

GENERAL OUTCOMES AND KEY COMPETENCES

General Outcomes:	Key competences
• Develop an understanding of Chemistry and its branches • Develop investigative skills about Chemistry • Demonstrate an understanding of the particulate nature of matter • Develop investigative skills about states of matter • Demonstrate an understanding of Experimental Techniques and its application in everyday life • Develop investigative skills in Experimental Techniques • Demonstrate an understanding of atoms, elements, molecules and compounds. • Develop investigative skills about the nature of substances. • Demonstrate an understanding of the importance, production, use, and effect on the environment of common elements and simple compounds	• Demonstrate the ability to measure time ,temperature, mass and volume • Show basic skills and knowledge in constructing balanced chemical equations with state symbols • Demonstrate investigative skills in Experimental Techniques

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
10.1 INTRODUCTION TO CHEMISTRY	10.1.1 Introduction to Chemistry	10.1.1.1 Describe Chemistry. 10.1.1.2 Classify the branches of chemistry	• The study of matter and their chemical changes • Branches such as: Analytical, Biochemistry, Inorganic, Physical and Organic	• <i>Classifying</i> of chemistry into its branches • <i>Identifying</i> different branches of chemistry	• <i>Asking</i> questions for more understanding • <i>Awareness</i> of chemistry branches
		10.1.1.3 Explain the importance of chemistry. 10.1.1.4 Describe the challenges of chemical industrial activities 10.1.1.5 Demonstrate an appreciation of safety in the laboratory.	• Improved life through manufacture of soaps, detergents, plastic, sugar, cement, paper, medicines, food production and other life necessities • Production of undesired harmful by-products. • Safety rules in the lab	• <i>Differentiating</i> chemistry from the other natural sciences	• <i>Appreciating</i> chemistry

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
10.2 THE PARTICULATE NATURE OF MATTER	10.2.1 Matter and the Kinetic theory	10.2.1.1 Describe matter	- Anything that has mass and occupies space - Atoms ,molecules ,ions	Classifying the basic units and states of matter	Appreciating the basic units of matter and its existence in three states
		10.2.1.2 Classify the basic units of matter 10.2.1.3 Classify the states of matter.	Kinetic theory: in terms of particle arrangement and movement. Solid, liquid, gas		
		10.2.1.4 Illustrate changes of states of matter. 10.2.1.5 Describe the absorption of heat and release of heat during changes of states of matter	- Changes of states such as melting, freezing, boiling, condensation, sublimation in terms of kinetic theory - Changing states of matter, exothermic-release of heat during a reaction, endothermic-absorption of heat during a reaction, include heating and cooling curves	Demonstrating the changes of states of matter Inferring data on absorption and release of heat during changes of states of matter	Applying changes of states of matter in everyday life

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
	10.2.2 Diffusion	10.2.2.1 Describe diffusion 10.2.2.2 Demonstrate diffusion in fluids 10.2.2.3 Describe the factors that affect the rate of diffusion	• Movement of particles from region of higher concentration to region of lower concentration • Liquids and gases (Brownian motion) • Factors of diffusion E.g. molecular mass, temperature, concentration	• Demonstrating the movement of particles in fluids • Comparing movement of particles in fluids and factors affecting their speed of movement	• Appreciating diffusion • Asking more questions to learn more • Fostering teamwork
10.3 EXPERIMENTAL TECHNIQUES	10.3.1 Measuring of quantities	10.3.1.1 Demonstrate how different quantities are measured. 10.3.1.2 Identify different measuring apparatus used in chemistry. 10.3.1.3 Identify various measuring instrument and other apparatus used in chemistry	• Quantities such as time, temperature, mass and volume • Measuring apparatus such as stopwatch or stop clock, thermometers, balances, burettes, pipettes, volumetric flask, measuring cylinder, and gas syringes • Other apparatus: spatula, stands and clamp, test-tubes, burners, , glass rods, evaporating dish, funnel beaker, conical flask etc.	• Demonstrating accurate measurement of values of various quantities • Identifying different measuring apparatus	• Applying safety rules in use of apparatus

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
	10.3.2 Criteria of purity	10.3.2.1 Describe the differences between a pure substance and a mixture	In terms of melting points and boiling points	Differentiating between melting points and boiling points	Appreciating purity of substances
		10.3.2.2 Demonstrate how to determine the purity of a substance	Sharp melting for pure substance and melting over a range of temperatures for a mixture.	Demonstrating determination of purity of substances Comparing pure and impure substances	
		10.3.2.3 Explain the importance of purity of a substance	Importance of purity in substances such as foodstuffs, medicines, drinks	Communicating the importance of purity in substances	Appreciating of the purity of substances

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
	10.3.3 Separating mixtures	10.3.3.1 Distinguish between physical and chemical changes 10.3.3.2 Demonstrate different methods of separating mixtures 10.3.3.3 Interpret simple paper chromatograms.	• In terms of mass changes, irreversibility/reversibility, chemical substance formed and energy involved. • Methods such as decantation, filtration, crystallisation, simple and fractional distillation, magnetism, chromatography, evaporation, sublimation, floatation, use of separating funnel and centrifugation • Uses such as R_f values and distances covered by components (restricted to paper chromatography)	• Analysing the components in the mixture • Identifying appropriate methods for separating different mixtures	• Applying separation techniques in everyday life

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
10.4 ATOMS, ELEMENTS, COMPOUNDS AND MOLECULES	10.4.1 Atomic structure and Periodic Table	10.4.1.1 Describe an atom and its structure.	As the smallest particle of an element which takes part in a chemical reaction. Structure: use Bohr model (nucleus at the centre surrounded by electron shells)	Communicating atoms, elements molecules and compounds Calculating relative atomic mass	Awareness of the atomic structure
		10.4.1.2 Describe the relative charges and approximate relative masses of protons, neutrons and electrons	- Charges as: +1,0,-1 - Masses as: 1, 1, 1/1840		
		10.4.1.3 Describe the proton (atomic) number and nucleon(mass) number and nuclide notation	As number of protons: Z, number of nucleons: A (protons + neutrons) and nuclide notation A_ZX	Calculating relative atomic mass	Asking more questions to learn more Fostering teamwork

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		10.4.1.4 Describe an element 10.4.1.5 Identify elements using their chemical symbols 10.4.1.6 Describe the basis of the Periodic Table	- As substance made up of same chemical atoms. - Symbols of the elements with atomic number 1 up to 20 and other common elements in the local environment - Group determined by valence electrons Period determined by number of shells	Communicating elements and the periodic table	Appreciating elements from the environment
		10.4.1.7 Describe isotopes 10.4.1.8 Calculate relative atomic mass of an element given the % abundances of isotopes and from mass spectrum.	- As atoms with same number of protons but different numbers of neutrons, including radioactive and non-radioactive isotopes - As sum of the products of the percentages and their mass numbers	Calculating relative atomic mass of an element	Asking more questions to learn more Fostering teamwork

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		10.4.1.9 Describe the use of radioactive isotopes 10.4.1.10 Demonstrate the build-up of electrons in shells	- Such as in medical treatment of cancer, industrial use as tracers - Electronic configuration of atoms (spdf configuration is NOT required)	Communicating the uses of isotopes	Awareness of the uses of isotopes
	10.4.2 Bonding	10.4.2.1 Describe a compound 10.4.2.2 Describe the formation of ions (radicals). 10.4.2.3 Describe the formation of ionic (electrovalent) bonds.	- As substance formed from two or more elements chemically combined - Cations by electron loss, anions by electron gain. - Electrovalent bonding as loss and gain of electrons between metallic and non-metallic atoms. Ionic bonds as electrostatic force between cations and anions. Such as NaCl, CaCl₂ and MgO	Classifying ionic compounds and covalent compounds	Appreciating the use of ionic compounds and covalent compounds

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		10.4.2.4 Describe the formation of covalent bonds	- Covalent bonding as sharing of electrons between non-metallic atoms. Covalent bonds as shared pairs of electrons. Such as H₂, Cl₂, H₂O, NH₃, CH₄, HCl, C₂H₆ - Such as double bonds in O₂, C₂H₄ and CO₂, Triple bond in N₂ and C₂H₂	Communicating the formation of covalent bonds Inferring the arrangements of simple multiple covalent molecules Demonstrating bond formation using models	Asking more questions to learn more Fostering teamwork
		10.4.2.5 Describe the electronic arrangement in simple multiple covalent molecules			
		10.4.2.6 Describe the uses of ionic and covalent compounds	- As refractory materials for ionic compounds (CaO) and polar and nonpolar solvents for covalent compounds. - As the smallest particle of an element or compound which exists independently.	Communicating the uses of ionic and covalent compounds	Asking more questions to learn more Fostering teamwork Appreciating ionic and covalent compounds
		10.4.2.7 Describe a molecule			

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		10.4.2.8 Describe valency and valence electrons. 10.4.2.9 Demonstrate how to deduce valency of an element.	- Valency as combining power of an atom or radical. Valence electrons as the number of electrons in the outer most shell. - From the formula of a compound, ionic charge, valence electrons.	Demonstrating the deducing of valency	Asking more questions to learn more Fostering teamwork
		10.4.2.10 Formulate chemical formulae of compounds. 10.4.2.11 Identify the differences in properties of ionic and covalent compounds.	- Using valency and chemical symbols of elements, charges on ions, models, relative numbers of atoms present, diagrammatic representation - Differences such as volatility, electrical conductivity, density, melting point, boiling point and basic units.	Formulating chemical formulae Differentiating chemical formulae Investigating properties of ionic and covalent compounds	Asking more questions to learn more Fostering teamwork Appreciating ionic and covalent compounds

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		10.4.2.12 Describe metallic bonding 10.4.2.13 Describe the electrical and thermal conductivity of metals	- As lattice of positive ions in a 'sea' of delocalised electrons - Electrical and thermal: due to free electron movement/delocalised electrons	Communicating metallic bonding and thermal conductivity	Asking more questions to learn more
	10.4.3 Macromolecules	10.4.3.1 Describe the giant covalent structures of graphite and diamond 10.4.3.2 Describe the uses of graphite and diamond in relation to their structures 10.4.3.3 Describe the macromolecular structure of silicon (IV) oxide(silicon dioxide) 10.4.3.4 Identify the similarities in properties between diamond and silicon dioxide	- Graphite as giant structures of carbon atoms arranged in hexagonal layers while diamond as a giant structure of carbon atoms arranged tetrahedrally. - Uses of graphite as a lubricant, pencil leads, electrodes and uses of diamond in cutting, jewellery - Macromolecular structure of silicon (IV) oxide(silicon dioxide): As oxygen atoms bonded to silicon atoms tetrahedrally despite the formula being SiO_2. - Similarities in properties between diamond and silicon dioxide: such as atoms held together by covalent bonds tetrahedrally.	Comparing the structures and uses of graphite, diamond and silicon dioxide. Comparing the properties of diamond and silicon dioxide.	Awareness of macromolecules . Appreciating the structures and uses of macromolecules Cooperating in group work.

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
	10.4.4 Chemical formulae and equations	10.4.4.1 Demonstrate how to construct word equations. 10.4.4.2 Formulate and balance chemical equations. 10.4.4.3 Construct net ionic equations from balanced chemical equations.	- Equation showing reactants and products separated by a full curled arrow ($\rightarrow$). - Number of atoms of each element being equal on both sides of the equation. Balancing can be done by inspection. Equations may include state symbols (s-solid, l – liquid, g – gas, aq – aqueous). - Only ionic aqueous reactants/products must be broken down into their respective ions then cancel out spectator ions to come up with net ionic equation.	Demonstrating construction of word equations Formulating balanced chemical and ionic equations.	Appreciating the conservation of matter.

GRADE 11

GENERAL OUTCOMES AND KEY COMPETENCES

General Outcomes:	Key Competences
• Demonstrate an understanding of acids, bases and salts. • Develop investigative skills about acids, bases and salt. • Demonstrate an understanding of the importance, production, use, and effect on the environment of acids, bases and salts. • Demonstrate an understanding of the Mole Concept • Develop investigative skills about quantitative analysis. • Demonstrate an understanding of chemical reactions and energy changes • Develop investigative skills about various types of reactions. • Demonstrate an understanding of the Periodic Table • Develop investigative skills about the Periodic Table	• Demonstrate the skills and knowledge in relating number of valence electrons to the Group number and the number of shells to the Period. • Demonstrate skills in classifying salts according to their solubility. • Demonstrate ability to classify oxides as acidic, basic, neutral and amphoteric. • Demonstrate ability to use tests in identifying aqueous cations, anions and gases. • Demonstrate basic skills and knowledge in calculating stoichiometric reacting moles. • Show ability to identify factors that affect rates of chemical reactions.

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
11.1 ACIDS, BASES AND SALTS	11.1.1 Characteristic properties of acids and bases	11.1.1.1 Describe acids, bases or alkalis in terms of ions they contain or produce in aqueous solution.	- Acid as compound that produces hydrogen ions as the only positively charged ions in aqueous solutions, Base generally as an oxide or hydroxide of a metal including ammonium hydroxide - Alkalis as soluble bases that produce hydroxide ions in aqueous solution as the only negatively charged ions.	<i>Identifying</i> acids and bases. <i>Investigating</i> the acidity and alkalinity of substances in everyday life	<i>Applying</i> the uses of acids and bases
		11.1.1.2 Describe the meaning of weak, strong, dilute and concentrated acids and alkalis	Strength as degree of ionisation, Concentration as the number of ions per volume of solution.		

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		11.1.1.3 Describe the P ^H scale	- As scale ranging from 0 to 14 showing the degree of acidity and alkalinity. - The P^H values: 7 for neutrality, below 7 for acidity and above 7 for alkalinity	<i>Identifying</i> acids and bases. <i>Investigating</i> the acidity and alkalinity of substances in everyday life	<i>Applying</i> the uses of acids and bases
		11.1.1.4 Describe neutrality, acidity and alkalinity in terms of P ^H value			
		11.1.1.5 Determine the P ^H value of a solution.	- Using universal indicator: different colours at different P^H values, Using P^H meter: precise values - Such as reactions with metals, bases, carbonates/bicarbonates and effect on indicators. - Such as reactions with acids and ammonium salts, effect on indicators. - Such as in controlling the acidity in the soil, treatment of indigestion, brushing teeth with toothpaste.		
		11.1.1.6 Demonstrate the characteristic properties of acids			
		11.1.1.7 Demonstrate the characteristic properties of bases			
11.1.1.8 Illustrate the importance of acid-base reactions					

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		11.1.1.9 State the uses of acids and bases.	Such as control of P^H in agriculture, making of soap, in car batteries	Identifying acids and bases. Investigating the acidity and alkalinity of substances in everyday life	Applying the uses of acids and bases
	11.1.2 Preparation of salts	11.1.2.1 Describe a salt 11.1.2.2 Classify salts according to their nature and solubility in water 11.1.2.3 Demonstrate the preparation of an insoluble salt. 11.1.2.4 Demonstrate the preparation soluble salts.	- As a compound formed when the hydrogen ions of an acid are fully or partially replaced by a metal or ammonium ions. Or a compound made of positive metallic/ammonium ions and any negative ion of an acid. - As acid, basic and normal salts. Solubility rules of salts - Using precipitation method and separated by filtration. E.g. Barium sulphate, Silver chloride - By reaction of acids with bases, suitable metals and carbonates/bicarbonates. Separated by crystallisation and filtration. E.g. Zinc sulphate, copper (II) sulphate	Classifying of salts Demonstrating the preparation of soluble and insoluble salts Differentiating hydrated and anhydrous salts Experimenting on preparation of salts	Awareness of salts Applying safety rules in preparation of salts

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		11.1.2.5 Demonstrate the preparation of ammonium, potassium and sodium salts. 11.1.2.6 Demonstrate the existence of hydrated salts and differentiate from anhydrous salts	- Using titration method (use of indicator for ease detection of end point) - Hydrated salts as salts containing water of crystallisation. Anhydrous salts as salts not containing water of crystallisation.	Classifying of salts Demonstrating the preparation of soluble and insoluble salts Differentiating hydrated and anhydrous salts Experimenting on preparation of salts	Awareness of salts Applying safety rules in preparation of salts
		11.1.2.7 Describe the behaviour of salts with reference to the atmosphere.	As hygroscopic, efflorescent, deliquescent.	Classifying of salts Demonstrating the preparation of soluble and insoluble salts Differentiating hydrated and anhydrous salts Experimenting on the behaviour of salts	Awareness of salts Applying safety rules in preparation of salts

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
	11.1.3 Types of oxides	11.1.3.1 Describe the various types of oxides.	- Acidic oxides as oxides with acidic properties such as SO_2 and CO_2. - Basic oxides as oxides with basic properties such as CaO and MgO. - Neutral oxides as oxides with neither acidic nor basic properties such as CO, H_2O. - Amphoteric oxides as oxides with both acidic and basic properties ZnO, Al_2O_3 and PbO.	Classifying different types of oxides	Awareness of different types of oxides. - Applying acid-base reactions
	11.1.4 Identification of ions and gases (Qualitative analysis)	11.1.4.1 Demonstrate the identity of aqueous cations and anion.	- Cations being aluminum, ammonium, calcium, copper (II), iron (II), iron (III), and zinc using aqueous sodium hydroxide and aqueous ammonia. - Anions being carbonate, chloride, iodide, nitrate and sulphate using various reagents. Refer to Qualitative notes	Observing and interpreting results of reactions of ions with different test reagents. Analysing chemical composition of salts.	Awareness about composition of salts

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		11.1.4.2 Demonstrate the identity of gases.	Gases being ammonia, carbon dioxide, chlorine, hydrogen, oxygen and sulphur dioxide. Refer to Qualitative notes	Identifying gases	Appreciating different types of gases.
11.2 THE MOLE CONCEPT	11.2.1 Relative masses	11.2.1.1 Describe Relative Atomic Mass and relative molecular mass. 11.2.1.2 Calculate the relative formula mass of a compound	- RAM as relative mass of an element's isotopes as compared to carbon-12 RMM as relative mass of a molecule as compared to carbon-12 - As the sum of the relative atomic masses of all the atoms in the compound.	Comparing the relative atomic masses and relative molecular masses Calculating relative molecular mass of compounds	Appreciating the relative atomic masses and the relative molecular masses
	11.2.2 The mole	11.2.2.1 Describe a mole. 11.2.2.2 Determine the physical masses (m) of any substance using the molar mass (Mr) and the physical volume (v) of any gas at r.t.p and vice versa.	- As number or quantities of particles e.g. atoms, ions, molecules, electrons equivalent to 6.02×10^{23} (Avogadro's constant) - Apply $n = \frac{m}{M_r}$ and $n = \frac{v}{V_m}$ where n = number of moles	Analysing chemical substances quantitatively	Applying mole concept Asking questions to learn more Awareness of the mole concept Fostering team work

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		11.2.2.3 Describe the relationship of Avogadro's law to reacting moles and volumes of gases at r.t.p and s.t.p. 11.2.2.4 Determine the concentration of a solution and apply dilution law.	- As Molar gas volume (V_m) of any gas at rtp is 24dm^3 or 22.4 dm^3 at stp. - Concentration as mol/dm^3 or g/dm^3. The number of moles of solute before dilution is the same as after dilution, $M_1 V_1 = M_2 V_2$.	Demonstrating acid-base titrations Problem solving in mole concept	Applying mole concept Asking questions to learn more Awareness of the mole concept Fostering team work
		11.2.2.5 Illustrate calculations involving stoichiometric reacting moles and volumes of gases and solutions.	Using molar mass and molar volume of a gas using the mole concept. (Questions on gas laws and conversions of gaseous volumes to different temperatures and pressures will not be required). Proportional stoichiometric masses and the given quantities	Problem solving in mole concept	Applying mole concept Asking questions to learn more Awareness of the mole concept Fostering team work

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		11.2.2.6 Describe and calculate the percentage yield in a reaction and the percentage purity of a substance 11.2.2.7 Determine limiting reagent in a given reaction	• % yield as actual amount obtained divided by theoretical amount x 100% • % purity as amount of substance divided by total amount of the mixture x 100% • Using proportional stoichiometric masses and the given quantities	• Problem solving in mole concept	• Applying mole concept • Asking questions to learn more • Awareness of the mole concept Fostering team work
		11.2.2.8 Demonstrate calculations involving different types of acid–base titration reactions.	• The different types of acid–base titration reactions: Using titration law	• Demonstrating acid-base titrations	• Applying mole concept • Asking questions to learn more • Awareness of the mole concept Fostering team work

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
	11.2.3 Empirical and Molecular formulae	11.2.3.1 Calculate the percentage composition of elements in a compound. 11.2.3.2 Determine the empirical formulae of a compound given the molecular formula	- Percentage composition of elements in a compound: As relative mass of element divided by relative formula mass of compound $\times 100\%$ - The empirical formulae by using atom ratios	Appreciating chemical composition of substances Applying chemical analysis Fostering team work. Asking more questions for better understanding.	Appreciating chemical composition of substances Applying chemical analysis Fostering team work. Asking more questions for better understanding.
		11.2.3.3 Determine the empirical and molecular formulae using percentage composition or masses.	Empirical formula: By calculating the number of moles of each component then converts to the simplest mole ratios to get empirical formula. Molecular formulae as a multiple of empirical formula given the relative molecular mass. $M.F = (E.F)_n$, where $n = RMM/REM$	Appreciating chemical composition of substances Applying chemical analysis Fostering team work. Asking more questions for better understanding.	Appreciating chemical composition of substances Applying chemical analysis Fostering team work. Asking more questions for better understanding.

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
11.3 CHEMICAL REACTIONS	11.3.1 Rates of chemical reactions	11.3.1.1 Describe rate of a chemical reaction. 11.3.1.2 Demonstrate the factors that affect the rates of chemical reactions 11.3.1.3 Interpret data on the rate of chemical reactions.	• As speed of a chemical reaction. • Factors such as temperature, concentration, surface area, catalyst, pressure, light. • Such as graphical representations for rate of chemical reactions.	• Demonstrating factors that control the rate of chemical reactions. • Comparing experimental results at different conditions	• Applying safety rules and the factors that affect the rate of chemical reactions. Awareness of slow and spontaneous reactions.
		11.3.1.4 Describe methods of controlling the rate of chemical reactions. 11.3.1.5 Describe the effect of a catalyst on the activation energy	• Made by either reducing or reducing the frequency of collisions between reacting particles such as explosions in flour mills/coal mines when ignited to surface area • Catalyst lowers the activation energy thus increasing the rate of a chemical reaction.	• Analysing and interpreting experimental results.	• Applying safety rules and the factors that affect the rate of chemical reactions. Awareness of slow and spontaneous reactions.

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
	11.3.2. Chemical equilibrium	11.3.2.1 Describe what chemical equilibrium is 11.3.2.2 Describe the effect on the position of equilibrium of a reaction upon changing conditions.	- Chemical equilibrium: as when the rate of the forward reaction equalises with rate of the backward reaction and the concentrations of the substances remain constant. It occurs in reversible reactions. - As changes in temperature, pressure, concentration. Apply Le Chatelier's Principle (candidates may not be required to state the principle) NB:Catalysts have no effect on the position of equilibrium	Communicating information in reversible reactions. Classifying reactions as reversible and irreversible. Comparing reversible reactions to irreversible reactions. Inferring the direction of a reaction based on relevant data.	Awareness of reversible reactions. Applying and appreciating the optimum conditions of a reaction. Asking more questions for better understanding.
	11.3.3 Redox reactions	11.3.3.1 Describe what oxidation and reduction is	Redox in terms of electron transfer, hydrogen/oxygen transfer, changes in oxidation state.	Experimenting with redox reactions.	Awareness of redox and non-redox reactions

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		11.3.3.2 Describe what redox reactions is. 11.3.3.3 Identify oxidizing and reducing agents in a reaction. 11.3.3.4 Demonstrate how to determine the oxidation number of an element with variable valency in a compound/ion. 11.3.3.5 Deduce a redox reaction using oxidation numbers. 11.3.3.6 Describe what non-redox reaction is	• Oxidizing agent as a reactant that gains electrons and/or reduces oxidation state • Reducing agent as a reactant that loses electrons and/or increases oxidation state • Determining oxidation numbers using standard rules. • As changes in oxidation numbers of reactants and products • Non-redox reaction: As reaction in which there is neither oxidation nor reduction involved.	• Observing colour changes. • Comparing oxidizing and reducing agents • Classifying redox and non-redox reactions • Planning an experiment to show the effects of an oxidizing and reducing agent	• Applying redox reactions • Appreciating redox reactions. • Participating actively in group activities • Applying safety rules when experimenting
		11.3.3.7 Identify the characteristics of oxidizing and reducing agents	• Oxidizing agents identified using potassium iodide solution as reducing agent in the presence of starch or starched potassium iodide paper. Reducing agents identified using acidified potassium dichromate or potassium permanganate as oxidizing agents and observe colour changes only. NB: No equations involving potassium dichromate and potassium permanganate will be required.	• Predicting the oxidation number of an element • Predicting a reaction as being redox or non-redox.	• Applying redox reactions • Appreciating redox reactions. • Participating actively in group activities • Applying safety rules when experimenting

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
	11.3.4 Energetics of reactions	11.3.4.1 Describe what endothermic and exothermic reactions are 11.3.4.2 Determine a reaction which is endothermic or exothermic.	- Types of Energetics of reactions: As energy in(endothermic) and energy out(exothermic) reactions - Identification of endothermic or exothermic reactions: Such as changes in enthalpy, energy level diagrams. Calculating ΔH using bond energies.	Experimenting with endothermic and exothermic reactions. Classifying reactions as endothermic and exothermic.	Applying endothermic and exothermic reactions Applying safety rules during experiment
		11.3.4.3 Describe endothermic and exothermic reactions in relation to bonds. 11.3.4.4 Identify activation energy for a catalysed and uncatalysed reaction on an energy level diagram 11.3.4.5 Explain the advantages and disadvantages of energy sources (fuels). 11.3.4.6 Describe the effects of the use of fuels on the environment	- Endothermic as bond breaking and exothermic as bond formation. - As in energy level representations: lower for catalysed and higher for uncatalysed. - Advantages and disadvantages of energy sources: Such as safety, cost of available reserves, renewable/non-renewable sources - Effects of the use of fuels on the environment: Such as pollution, greenhouse effect (global warming).	Classifying reactions as endothermic and exothermic. Investigating the effects of fuels on the environment	Applying endothermic and exothermic reactions Applying safety rules during experiment Applying information on energies of reactions in everyday life.

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		11.3.4.7 Describe the use of silver halide in photography 11.3.4.8 Describe respiration and photosynthesis in terms of energy changes	• Use of silver halide in photography: As reduction of silver ions to metallic silver by absorption of light.(endothermic reaction) • Respiration as exothermic process between oxygen and glucose producing carbon dioxide and water Photosynthesis as endothermic process between water and carbon dioxide through absorption of light producing glucose and oxygen.	• Planning an investigation to show that respiration and photosynthesis involve energy changes. • Communicating use of halides in photography	• Applying safety rules during experiment • Applying information on energies of reactions in everyday life.
		11.3.4.9 Describe use of radioactive isotope in relation to energy changes. 11.3.4.10 Explain batteries as convenient source of electrical energy.	• The use of radioactive isotope: As a source of nuclear energy. • The convenient source of electrical energy: such as batteries as they are potable.	• Investigating the effects of fuels on the environment • Investigating to show that batteries are a source of electrical energy.	• Applying information on energies of reactions in everyday life.

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
11.4 THE PERIODIC TABLE	11.4.1 Groups and Periods	11.4.1.1 Describe the Period Table 11.4.1.2 Identify vertical columns and horizontal rows. 11.4.1.3 Demonstrate how to use the Periodic Table to classify elements	- As a tool for classifying elements. - Vertical columns as Groups and horizontal rows as Periods - As metallic and non-metallic	Identifying of vertical columns and horizontal rows of the periodic table. Classifying elements as metallic and non-metals	Appreciating the Periodic Table Applying the classification of elements
	11.4.2 Groups and Periodic trends	11.4.2.1 Describe trends in various Groups given information about the elements 11.4.2.2 Describe the physical and chemical properties of elements in Group I, II, VII and VIII. 11.4.2.3 Describe the importance of halogens	- As chemical relativity of group I, II, and VII, elements - Properties such as solubility, effect of heat on compounds, melting points, boiling points and displacement reactions. For Group VII consider atomicity, colour changes, changes in physical states, for Group I including description as a collection of soft metals. - Such as fluoride in toothpaste, chlorine in water treatment, antiseptic, bromide in photographic film	Identifying characteristics of representative elements from Groups. Classifying elements according to their Groups and Periods	Awareness of elements and their positions on the Periodic Table Appreciating positioning of elements on the Periodic Table Appreciating the uses of elements on the Periodic Table

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		11.4.2.4 Describe the harmful effects of halides. 11.4.2.5 Describe the use of the noble gases in providing an inert atmosphere	• Such as drugs, pesticides, CFCs in ozone layer depletion (CFCs) • The significance of their non-reactivity in providing an inert atmosphere. Such as argon in electrical lamps, helium in balloons	• Communicating harmful effects of halides and uses of noble gases	• Awareness of harmfulness of halides • Appreciating of uses of noble gases
	11.4.3 Transition metals	11.4.3.1 Describe transition metals. 11.4.3.2 Describe general properties of transition metals. 11.4.3.3 Describe the uses of transition metals	• As a block elements between Group II and Group III of the Periodic Table • As variable valencies, high densities, high melting points, form coloured compounds, catalysts. Note: Electronic configuration of transition metals will not be required • Uses such as catalysts, alloys, engineering materials NB: Heavy metals are no longer used to make paint for health reasons	• Investigating the physical and chemical properties of transition elements. • Identifying transition metals	• Appreciating transition metals

GRADE 12

GENERAL OUTCOMES AND KEY COMPETENCES

General Outcomes:	Key competences:
• Demonstrate an understanding of Electricity and chemistry • Develop investigative skills about conductivity • Demonstrate an understanding of metals • Develop investigative skills about some properties and uses of metals. • Demonstrate an understanding of Non- metals. • Develop investigative skills about some industrial uses of non-metals • Demonstrate an understanding of Organic Chemistry	• Demonstrate ability to determine the reactivity series of metals • Demonstrate ability to prepare and test gases • Demonstrate ability to classify conductors,non-conductors,electrolytes and non-electrolytes • Show understanding of common pollutants of land , water ,and air • Demonstrate ability to identify organic compounds

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
12.1 CHEMISTRY AND ELECTRICITY	12.1.1. Conductors	12.1.1.1 Classify conductors and non- conductors	Conductors being metals such as copper, aluminium, silver and Non-conductors being non-metals such as sulphur, phosphorus, except carbon in form of graphite.	Classifying conductors and non conductors. Planning an experiment to investigate conductors and non conductors. Experiment to conductors and non conductors.	Appreciating conductivity of substances. Participating in class work actively Applying safety rules when experimenting
	12.1.2 Electrolysis	12.1.2.1 Classify electrolytes and non-electrolytes 12.1.2.2 Describe what electrolysis is	- Difference between electrolytes and non-electrolytes: Electrolytes as ionic compounds and non-electrolytes as covalent compounds. - Electrolysis: As decomposition of electrolyte using electricity in an electrolytic cell.	Classifying electrolytes and non electrolytes Formatting a hypothesis about the products at the electrodes during electrolysis.	Asking more questions for better understanding Awareness of the application of electrolysis in everyday life.

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		12.1.2.3 Describe the products at the electrodes during electrolysis of molten binary ionic compounds. 12.1.2.4 Describe the products at the electrodes during electrolysis of aqueous ionic solutions.	• Products by electrolysis of molten binary ionic compounds: metals at the cathode by reduction, non-metals at the anode by oxidation • Products by electrolysis of aqueous ionic solutions: Refer to selective discharge of ions given conditions. Electrolytes should include molten lead (II) bromide, molten aluminium oxide, dilute sulphuric acid (acidified water), concentrated hydrochloric acid, concentrated aqueous sodium chloride (brine) using carbon electrodes, and aqueous copper (II) sulphate using carbon and copper electrodes.	• Classifying electrolytes and non electrolytes • Formatting a hypothesis about the products at the electrodes during electrolysis.	• Asking more questions for better understanding • Awareness of the application of electrolysis in everyday life. • Applying safety rules when experimenting. • Participating actively in group work.

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		12.1.2.5 Describe the industrial applications of electrolysis. 12.1.2.6 Calculate the quantity of electrolytic products.	- Applications of electrolysis: Such as extraction of aluminium from its oxide, copper refinery and electroplating. - The quantity of electrolytic products: Using Faradays laws	Communicating the industrial application of electrolysis. Calculating the quantities of electrolytic products.	Awareness of the application of electrolysis in everyday life. Applying safety rules when experimenting. Participating actively in group work.
	12.1.3 Simple cells (chemical cell)	12.1.3.1 Describe what a chemical cell is 12.1.3.2 Compare electrolytic cells and simple cells	- A chemical cell: Two different metals connected together and dipped in an electrolyte to produce electricity. - Types of cells: Similarities such as oxidation at the anode and reduction at the cathode. Differences such as cathode being negative in electrolytic cell while positive in simple cell and vice versa for the anode. Simple cell must use two different electrodes while electrolytic cell can use any.	Comparing electrolytic and simple cells Classifying the electrolytic cells and simple cells in the chart	- Awareness of the two types of cells. - Asking more questions for better understanding

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
12.2 METALS	12.2.1 General properties of a metals	12.2.1.1 Describe diagrammatic representations of pure metals 12.2.1.2 Describe the physical properties of metal 12.2.1.3 Describe the chemical properties of metals	- Similar nuclei positive ions in a 'sea' of delocalised electrons. - Similar In terms of density, melting points, boiling points, appearance - All metals are electropositive as illustrated in the reaction with air, water / steam, dilute non- oxidizing acids, aqueous solutions of other metal ions.	Identifying properties of metals.	Appreciating metals
	12.2.2 Reactivity and Electro Chemical Series	12.2.2.1 Describe the reactivity series of metals	As arrangement of metals in the order of either their increasing or decreasing order of reactivity as being potassium, sodium, calcium, magnesium, aluminium, zinc, iron, lead, (hydrogen), copper and silver	Comparing methods of extracting metals.	Awareness of methods of extracting metals and dangers some metals pose.

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		12.2.2.2. Explain the apparent non reactivity of aluminium.	Due to the presence of adhesive oxide/coat. Reactivity of aluminium due to adhesive coat	Demonstrating reactivity of aluminium and order of reactivity	Appreciating of aluminium
		12.2.2.3 Demonstrate an order of reactivity.	From a set of experimental results Such as reduction of oxides of metals by other metals.		
		12.2.2.4 Describe the effects of heat on hydroxides, carbonates, nitrates of metals and ammonium compounds.	As related to the reactivity/stability of the metallic ion present in the compound. Compounds of more reactive metals difficulty to decompose while compounds of less reactive metals easily decompose.	Demonstrating effects of heat on salts	Asking questions to learn more Awareness of the heat on salts Fostering team work

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		12.2.2.5 Describe the extraction of copper, iron and zinc from their ores. 12.2.2.6 Describe the uses of copper, iron, zinc and aluminium 12.2.2.7 Explain the harmful effects of some metals.	• Chemical reduction. Chemical reducing agents being Carbon, carbon monoxide, and hydrogen. • Such as electrical wires, construction, aircraft parts. • Such as lead poisoning (brain damaging), sodium ions in raising high blood pressure, alzheimeria by aluminium	• Comparing methods of extracting metals.	• Asking questions to learn more • Awareness of the importance of metals • Fostering team work
	12.2.3 Alloys	12.2.3.1 Describe alloys. 12.2.3.2 Describe diagrammatic representations of alloys.	• As mixture of two or metals/carbon such as steel, brass, bronze • Different nuclei positive ions in a 'sea' of delocalised electrons	• Identifying characteristics of alloys • Comparing structures of alloys and pure metals.	• Appreciating alloys.

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		12.2.3.3 Explain the advantages of using alloys over pure metals.	Such as alloys exhibiting better properties compared to a pure metal (conductor, strength, weight ratio, hardness).	Identifying characteristics of alloys Comparing structures of alloys and pure metals.	Appreciating alloys.
	12.2.4 Corrosion	12.2.3.4 Identify common uses of alloys 12.2.4.1 Describe corrosion 12.2.4.2 Relate corrosion to the reactivity of metals. 12.2.4.3 Describe different methods of preventing corrosion.	- Such as cutlery, food packaging, aircraft. - As chemical wearing of metals resulting from attack by atmospheric oxygen in presence of moisture. - As more reactive metals easily corrode while less reactive metals do not easily corrode. - Such as sacrificial protection, painting, greasing/oiling, alloying and galvanising.	Identifying corrosion. Applying methods of reducing corrosion. Relating sacrificial protection methods to reactivity series.	Appreciating ways of minimizing corrosion.

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
12.3 NON - METALS	12.3.1 General properties of non-metals.	12.3.1.1 Describe the physical and chemical properties of non-metals.	In terms of density, melting points, boiling points, oxidizing agent (electronegative elements)	Identifying the physical and chemical properties of non-metals	Appreciating non-metals.
	12.3.2. Hydrogen	12.3.2.1. Demonstrate the laboratory preparation, collection and test for hydrogen. 12.3.2.2 Describe the physical and chemical properties of hydrogen 12.3.2.3 Describe industrial manufacture of hydrogen.	- By action of moderate reactive metals on water/steam and dilute acids and collect by upward delivery method, puts out a lighted splint with a 'pop' sound. - In terms of colour, odour, density/"weight", solubility and chemical (effect on litmus, inflammability, combustion)(COWSLIPS) - By cracking, electrolysis of water (brine) and from natural gas	Demonstrating laboratory preparation of hydrogen.	Appreciating physical and chemical properties of hydrogen and its uses.

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		12.3.2.4 Describe the uses of hydrogen.	Such as reducing agent, fuel for rockets, manufacturing ammonia and margarine, balloons filler, welding.	Communicating the uses of hydrogen	Appreciating physical and chemical properties of hydrogen and its uses.
	12.3.3. Oxygen	12.3.3.1 Demonstrate the laboratory preparation, collection and test for oxygen. 12.3.3.2 Describe the physical and chemical properties of oxygen. 12.3.3.3 Describe the industrial manufacture of oxygen.	- By catalytic decomposition of hydrogen peroxide and thermal catalytic decomposition of potassium chlorate, collected above water and re-lights the glowing splint - Such as colour, odour, solubility, combustion - By fractional distillation of liquid air	Demonstrating laboratory preparation of oxygen. Observing the reaction. Communicating the uses of oxygen	Appreciating physical and chemical properties of oxygen and its uses.

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		12.3.3.4 Describe the uses of hydrogen.	Such as reducing agent, fuel for rockets, manufacturing ammonia and margarine, balloons filler, welding.	Communicating the uses of hydrogen	Appreciating physical and chemical properties of hydrogen and its uses.
		12.3.3.5 Demonstrate the laboratory preparation, collection and test for oxygen. 12.3.3.6 Describe the physical and chemical properties of oxygen. 12.3.3.7 Describe the industrial manufacture of oxygen.	- By catalytic decomposition of hydrogen peroxide and thermal catalytic decomposition of potassium chlorate, collected above water and re-lights the glowing splint - Such as colour, odour, solubility, combustion - By fractional distillation of liquid air	Demonstrating laboratory preparation of oxygen. Observing the reaction. Communicating the uses of oxygen	Appreciating physical and chemical properties of oxygen and its uses.

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		12.3.3.8 Describe the uses of oxygen in industry and in natural processes. 12.3.3.9 Explain the importance of ozone layer and dangers of its depletion.	• Such as burning, welding, in blast furnace and respiration • It traps radiation, if depleted by CFCs causes skin cancer, respiratory diseases	• Communicating the uses of oxygen	• Appreciating uses of oxygen .
		12.3.3.10 Demonstrate the chemical test for water. 12.3.3.11 Describe the importance of water.	• Using white anhydrous copper (II) sulphate which turns blue. • For laundry, drinking, as solvent.	• Demonstrating the chemical testing of water • Communicating the importance	• Appreciating importance of water

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
	12.3.4 Nitrogen	12.3.4.1 Describe industrial manufacture of nitrogen.	By fractional distillation of liquid air	Demonstrating laboratory preparation of ammonia.	Appreciating physical and chemical properties of nitrogen and ammonia and their uses.
		12.3.4.2 Explain the characteristics and importance of Nitrogen as a gas.	As non- reactive insoluble gas hence used as refrigerant, food packaging. Manufacture of ammonia gas.	Observing colour changes.	
		12.3.4.3 Demonstrate the preparation collection and test for ammonia in the laboratory	Action of a base on ammonium salt and collected by upward delivery method, turns damp red litmus paper blue.	Demonstrating laboratory preparation of ammonia.	Appreciating physical and chemical properties of nitrogen and ammonia and their uses.
		12.3.4.4 Describe the manufacture of ammonia.	Haber Process (Temperature, catalyst, pressure (Haber process)).	Observing colour changes.	

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		12.3.4.5 Describe the physical and chemical properties of ammonia.	In terms of colour, odour, density/"weight", solubility and as reducing agent, a base/alkali, a complexing reagent.	Demonstrating laboratory preparation of ammonia. Observing colour changes.	Appreciating physical and chemical properties of nitrogen and ammonia and their uses.
		12.3.4.6 Describe the thermal dissociation of ammonium salts.	Such as ammonium chloride, ammonium nitrate, ammonium carbonate.		
		12.3.4.7 Describe the uses ammonia	- In manufacture of fertilisers, explosives, nitric acid - by Ostwald Process		
		12.3.4.8 Describe the manufacture of nitric acid	- Nitrogen for growth. Include Phosphorous for root development and potassium for seed formation (NPK). - Such as eutrophication and acidic soils		
		12.3.4.9 Explain the importance of nitrogenous fertilizers			
		12.3.4.10 Describe the effects of nitrogenous fertilisers on the environment			

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
	12.3.5. Chlorine	12.3.5.1 Demonstrate the laboratory preparation, collection and test for chlorine gas.	By action of hot concentrated Hydrochloric acid on manganese (IV) oxide, collected by downward delivery method, turns damp blue litmus paper red and then bleaches it	Experimenting the laboratory preparation of chlorine. Observing colour changes during the preparation of chloride. Investigating the physical and chemical properties of chloride gas.	Awareness of physical and chemical properties of chlorine and its uses. Applying safety rules during experiments
		12.3.5.2 Describe the physical and chemical properties of chlorine gas	Physical and chemical properties of chlorine gas Such: as colour, odour, density, solubility, poisonous. Reactions with Iron, non-metals (H₂, S, O₂, P), sulphur dioxide, Iron(II)salts and halides.		
		12.3.5.3 Describe the uses of chlorine.	Uses of chlorine For sterilizing water, manufacture of PVC, HCl and in bleaching agents.	Communicating uses and industrial manufacture of chlorine	Awareness of physical and chemical properties of chlorine and its uses.
		12.3.5.4 Describe the industrial manufacture of chlorine.	Manufacture of chlorine: By the electrolysis of brine (NaCl_(aq))		

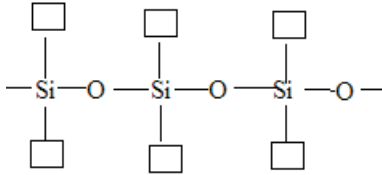
TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		12.3.5.5 Demonstrate the method for preparation, collection and test for hydrogen chloride gas	Preparation of hydrogen chloride: By action of concentrated sulphuric acid on solid metallic chlorides, collected by downward delivery method, react with ammonia to form white smoke.	Experimenting the preparation of hydrogen chloride gas. Investigating the physical and chemical properties of hydrogen chloride gas.	Awareness of physical and chemical properties of hydrogen chloride and its uses. Applying safety rules during experiments
		12.3.5.6 Describe the physical and chemical properties of hydrogen chloride gas	In terms colour, odour, density, solubility and poisonous. Reactions with ammonia and water		
		12.3.5.7 Demonstrate the method for preparation of hydrochloric acid.	By dissolving hydrogen chloride gas in water	Experimenting the preparation of hydrogen acid and its reactions.	Applying safety rules during experiments
		12.3.5.8 Describe the reactions of dilute hydrochloric acid.	Such as reaction with alkalis, metals, carbonates, ammonia and silver nitrate.		

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
	12.3.6 Sulphur	12.3.6.1 Describe the formation of sulphur dioxide.	- The formation of sulphur dioxide: By combustion of sulphur, fossil fuels - Laboratory preparation, collection and test for sulphur dioxide: By action of warm dilute acids on sulphites, collected by downward delivery, turns acidified potassium dichromate (VI) green/decolourises purple potassium manganate (VII).	Experimenting laboratory preparation of sulphur dioxide Observing colour changes during the preparation of sulphur dioxide	Awareness of the physical and chemical properties of sulphur dioxide and its uses. Applying safety rules when experimenting
		12.3.6.3 Describe the physical and chemical properties of sulphur dioxide	The physical and chemical properties of sulphur dioxide: In terms of colour, odour, density, solubility, poisonous. Reaction with water, action on indicators and as a reducing agent E.g. turns acidified potassium dichromate (VI) green/decolourises purple potassium manganate (VII).	Communicating information on properties of sulphur dioxide and its everyday life. Investigating properties of sulphur dioxide	Awareness of the physical and chemical properties of sulphur dioxide and its uses. Applying safety rules when experimenting

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		12.3.6.4 Describe the uses of sulphur dioxide	• Uses of sulphur dioxide: As food preservative, bleaching wood pulp for paper making, manufacture of sulphuric acid • Manufacture of sulphuric acid: By Contact Process (catalyst, temperature) • Uses of sulphuric acid: Such as in explosives, as drying agent, making of soaps, fertilisers	• Investigating uses of sulphur dioxide • Communicating the industrial manufacture of sulphuric acid	• Awareness of the physical and chemical properties of sulphur dioxide and its uses. • Applying safety rules when experimenting
		12.3.6.5 Describe the industrial manufacture of sulphuric acid.			
		12.3.6.6 Describe the uses of sulphuric acid.			
	12.3.7 Carbon and carbonates	12.3.7.1 Describe allotropes	• As different forms of an element existing in the same physical state. • In terms crystalline and non-crystalline allotropes of carbon.	• Investigating properties of allotropes of carbon	
		12.3.7.2 Describe the physical properties of the allotropes of carbon.			

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		12.3.7.3 Describe the formation and properties of carbon monoxide.	By incomplete combustion of carbon and carbon compounds, reduction of carbon dioxide by carbon. In terms of colour, odour, density, solubility, poisonous. Reacts as reducing agent.	Demonstrating laboratory preparation of carbon dioxide. Observing colour changes.	Appreciating physical and chemical properties of carbon dioxide and limestone and their uses.
		12.3.7.4 Demonstrate the laboratory preparation, collection and the test for carbon dioxide.	By reaction of dilute acids with carbonates or bicarbonates, collected by downward delivery method/ above water, forms white precipitate with limewater.		

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		12.3.7.5 Describe the physical and chemical properties of carbon dioxide.	In terms of colour, odour, density, solubility. Reactions with limewater/alkalis, water and carbon.	Demonstrating laboratory preparation of carbon dioxide. Observing colour changes.	Awareness of Global warming Appreciating physical and chemical properties of carbon dioxide and limestone and their uses.
		12.3.7.6 Describe the uses of carbon dioxide.	Such as in fire extinguishers, carbonated drinks, dry ice, baking, photosynthesis		
		12.3.7.7 Describe the manufacture of lime from limestone.	By thermal dissociation of limestone		
		12.3.7.8 Describe the uses of lime and slaked lime.	Such as in neutralizing acidic soils, lime as a drying agent for ammonia.		
		12.3.7.9 Describe the uses of limestone.	Such as in manufacturing of lime, cement, glass, iron.		
		12.3.7.10 Describe the greenhouse effect	As global warming due to increase of carbon dioxide in the atmosphere		

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
	12.3.8 Silicon	12.3.8.1 Describe the properties of silicon. 12.3.8.2 Describe the use of silicon. 12.3.8.3 Describe what silicones are.	- The properties of silicon: As a metalloid - Use of silicon: Used in semi-conductors Such as transistors, diodes and capacitors. - Silicones: As macromolecules that exist as oils, waxes or plastics and their structures represented as: 	Communicating the properties of silicon and its uses Comparing the fire resistance of macromolecules.	Awareness of the use of silicon in everyday life. Cooperating in group work.
		12.3.8.4 Compare the fire resistance of siliconeplastics to carbon based macromolecules	The nature of silicones: With reference to nature of combustion products, silicones produce silicon dioxide (sand) while organic based macromolecules produce carbon dioxide.	Comparing fire resistance of silicone to carbon based molecules	Awareness of the use of silicon in everyday life. Cooperating in group work.

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		12.3.8.5 Describe the uses of silicon dioxide (sand)	Uses of silicon dioxide: Such as in making glass, as fire extinguisher, in iron extraction.	Communicating the uses of silicon dioxide(sand)	Awareness of the use of silicon in everyday life. Cooperating in group work.
12.4 ORGANIC CHEMISTRY	12.4.1 Saturated and unsaturated Hydrocarbons	12.4.1.1 Describe an organic compound.	As a compound of carbon other than oxides and carbonates	Identifying alkanes and alkenes.	Appreciating economic values of alkanes and alkenes.
		12.4.1.2 Describe hydrocarbon	As a binary compound of carbon and hydrogen.	Comparing properties of alkanes and alkenes	Awareness of organic compounds.
		12.4.1.3 Illustrate and name the structures of the aliphatic alkanes up to five carbon atoms.	Involve concept of catenation (Chain), use the general formula C_nH_{2n+2}, Named by IUPAC system, all should end with <i>ane</i>, Involve concept of catenation (Chain), use the general formula C_nH_{2n+2}, Named by IUPAC system, all should end with <i>ane</i>,		

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		12.4.1.4 Demonstrate the structures of isomers and their names. 12.4.1.5 Describe fractional distillation of petroleum (crude oil) 12.4.1.6 Describe the uses of the fractions of crude oil 12.4.1.7 Describe the chemical properties of alkanes. 12.4.1.8 Account for the apparent non reactivity of alkanes as compared to other organic compounds. 12.4.1.9 Illustrate unsaturation in alkenes.	• Use idea of branched (side chains) and unbranched butane and pentane and nomenclature follows IUPAC system. • As different fractions of crude oil collected at different boiling temperatures. • Such as domestic fuel, road construction. NB: leaded fuel is no longer recommended due to harmful effects • Such as combustion, cracking, substitution, steam reforming. • Lack of a specific site of chemical attack (functional group) and they are saturated. • Using the concept of catenation and models	• Identifying alkanes and alkenes. • Comparing properties of alkanes and alkenes • Observing colour changes.	• Appreciating economic values of alkanes and alkenes. • Awareness of organic compounds. • Identifying alkanes and alkenes. • Comparing properties of alkanes and alkenes • Observing colour changes.

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		12.4.1.10 Describe and name the structures of the alkenes up to 5 carbon atoms. 12.4.1.11 Demonstrate the structures of isomers of alkenes. 12.4.1.12 Describe the chemical properties of alkenes.	• Use the concept of catenation and the general formula C_nH_{2n}. Structures must contain one carbon to carbon double bond. Named using the IUPAC system all should end with- <i>ene</i>. • Using the unbranched structures of butene and pentene (positional isomers). • Such as combustion, addition reactions (hydrogenation, hydration, hydrohalogenation, halogenation, addition polymerisation).	• Identifying alkanes and alkenes. • Comparing properties of alkanes and alkenes • Observing colour changes.	• Appreciating economic values of alkanes and alkenes. • Awareness of organic compounds. •
		12.4.1.13 Illustrate the differences and similarities between saturated and unsaturated Hydrocarbons. 12.4.1.14 Describe the chemical tests for unsaturated hydrocarbons (alkenes) 12.4.1.15 Describe the uses of alkenes.	• Using structures and bromine solution to distinguish between saturated and unsaturated hydrocarbons. • As alkenes decolourise bromine solution rapidly. • As in formation of polymers (Petrochemical industries)	• Identifying alkanes and alkenes. • Comparing properties of alkanes and alkenes • Observing colour changes	• Appreciating economic values of alkanes and alkenes. • Awareness of organic compounds.

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
	12.4.2 Alcohols (Alkanols)	12.4.2.1 Describe the chemical composition of an alcohol.	- As an organic compound with a hydroxyl group with general formula $C_nH_{2n+1}OH$ - Using concept of catenation (Chain). Named following IUPAC nomenclature and all should end with- <i>ol</i>).	Identifying composition of alcohols Comparing structures of alcohols	Appreciating economic values of alcohols. Awareness of organic compounds.
		12.4.2.2 Describe and name structures of primary alcohols up to five carbon atoms.			
		12.4.2.3 Demonstrate isomerism in alcohols	Using branched and unbranched and positional isomers of propanol, butanol and pentanol.	Identifying uses of alcohols Comparing properties of alcohols.	Appreciating economic values of alcohols. Awareness of organic compounds.
		12.4.2.4 Describe the formation of alcohols.	By hydration of alkenes, hydrolysis of esters and fermentation for ethanol.		
		12.4.2.5 Describe the chemical properties of alcohols	Such as combustion, esterification, dehydration and oxidation		
		12.4.2.6 Describe the uses of alcohols	Uses such as fuel, antiseptic, organic solvent, alcoholic beverages		

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
	12.4.3 Carboxylic acids (alkanoic acids)	12.4.3.1 Describe and name structures of carboxylic acids up to five carbon atoms.	Using concept of catenation (Chain), organic compounds with a carboxylic group (COOH), general formula $C_nH_{2n+1}COOH$, all should end with- oic acid.	Identifying structures of carboxylic acids.	Appreciating the properties and economic uses of carboxylic acids.
		12.4.3.2 Describe the formation of carboxylic acids	By the oxidation of alcohols and hydrolysis of esters		
		12.4.3.3 Demonstrate the chemical properties of carboxylic acids.	Such as reaction with bases, carbonates, metals and alcohols (esterification).	Demonstrating the chemical properties of carboxylic acids	Appreciating the properties and economic uses of carboxylic acids.
		12.4.3.4 Describe the uses of carboxylic acids	Such as formation of esters.		

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
	12.4.4 Esters (Alkanoates)	12.4.4.1 Describe and name the structures of esters up to five carbon atoms.	- Using the concept of catenation (Chain), Organic compounds with an ester link $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{O}- \end{array}$ and all should end with <i>-oate</i>. - Such as combustion and hydrolysis.	Identifying structures and characteristic properties of esters.	Appreciating the properties and economic uses of esters.
		12.4.4.2 Describe the chemical properties of esters			
		12.4.4.3 Describe the uses of esters and relate the uses to properties.	Such as in perfumes, food flavourants because of having pleasant smell.	Describing the chemical properties of esters	Appreciating the properties and economic uses of esters.
	12.4.5 Homologous series	12.4.5.1 Describe homologous series	As a collection of organic compounds belonging to the same family with the same general formula (consider alkanes, alkenes, alcohols, acids, esters).	Identifying different homologous series.	Awareness of homologous series.
		12.4.5.2 Describe the general characteristics of homologues (members).	Such as members of each homologous series have the same general formula and similar chemical properties. Physical properties (states, melting point, boiling point, density, solubility) of members show gradual changes as molecular mass changes. Adjacent members differ by CH_2 and have a general method of preparing members.		

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
	12.4.6 Macromolecules (Polymers)	12.4.6.1 Describe macromolecules (polymers)	- As giant molecules formed by combination of many small molecules (monomers). - As human made giant molecules (polymers). - By addition polymerisation E.g. polyethene, polyvinylchloride, polypropene, polystyrene. - As thermoplastics and Thermosets	Classifying macromolecules Identifying linkages in different macromolecules	Awareness of polymers. Appreciating economic use of polymers.
		12.4.6.2 Describe synthetic macromolecules.			
		12.4.6.3 Describe the formation of polyalkenes.			
		12.4.6.4 Classify plastics			
		12.4.6.5 Describe the formation of nylon and Terylene.	By condensation polymerisation, Nylon: from a diamine and dioic acid structures represented as: Terylene Nylon 	Classifying macromolecules Identifying linkages in different macromolecules	Awareness of polymers. Appreciating economic use of polymers.

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		12.4.6.6 Differentiate between the structure of Nylon and Terylene.	- Terylene: from diol and dioic acid. Structures represented as: Nylon as polyamide and Terylene as polyester. - Plastics used as in carrier bags, buckets, pipes - Nylon and terylene as in clothing, tents, strings, ropes.	Classifying macromolecules Identifying linkages in different macromolecules	Awareness of polymers. Appreciating economic use of polymers.
		12.4.6.7 Describe typical uses of plastics and synthetic fibres.			
		12.4.6.8 Describe the biodegradability of synthetic fibres.	- As non-biodegradable (cannot be broken down by microorganisms) - Such as Carbohydrates, proteins and fats (lipids).	Classifying macromolecules Identifying linkages in different macromolecules	Awareness of polymers. Appreciating economic use of polymers.
		12.4.6.9 Describe natural macromolecules			

TOPIC	SUB TOPIC	SPECIFIC OUTCOMES	CONTENT		
			KNOWLEDGE	SKILLS	VALUES
		12.4.6.10 Describe composition of carbohydrates	- Carbohydrates contain carbon, hydrogen and oxygen in the form $C_xH_{2y}O_y$ where x is a multiple of six. - In starch – glycosidic, $-O-\square-O-\square-$ Proteins – amide, fats – ester linkages - Such as difference and similarities between nylon and proteins. Terylene and fats.	Classifying macromolecules Identifying linkages in different macromolecules	Awareness of polymers. Appreciating economic use of polymers
		12.4.6.11 Identify linkages in starch, proteins and fats			
		12.4.6.12 Relate linkages in synthetic and natural polymers.			
		12.4.6.13 Describe hydrolysis of fats (saponification).	- As formation of soaps and glycerine (glycerol). - Using chromatography to identify the amino acids from proteins, simple sugars from starch.	Classifying macromolecules Identifying linkages in different macromolecules	Awareness of polymers. Appreciating economic use of polymers
		12.4.6.14 Identify the products of the hydrolysis of starch and proteins.			

CHEMISTRY PRACTICAL DATA

The following points should be considered during practical in chemistry:

- (i) The student should have the knowledge of volumetric analysis in relation to one set of titrations. The student is expected to comprehend acid-alkali titrations using ordinary methyl orange, screened methyl orange, phenolphthalein or any other suitable indicator. Other titrations using different reagents may be set as well e.g. redox titration.
- (ii) Other experiments involving the determinations of quantity, temperature change and rates of reactions are necessary. Experiments of this nature will rely on the use of ordinary apparatus in the laboratory.
- (iii) Experiments involving identification of an unknown substance or mixture could be set. A learner is expected to observe and investigate the expected outcome. This may comprise elementary chromatography and simple tests for oxidising and reducing agents. Detailed analysis is not necessary but a learner is expected to have the knowledge of the reactions of the cations with aqueous sodium hydroxide and aqueous ammonia which should include elementary cations like aluminium, ammonium, calcium, copper(II), iron (II), iron (III) and zinc.

A learner should also carry out the tests for the anions such as carbonate, chloride, iodide, nitrate and sulphate. Chemical tests for gases which should include ammonia, carbon dioxide, chlorine, hydrogen, oxygen and sulphur dioxide.

Organic substances and ions not mentioned above may be included in the practical sessions. A learner is expected to have sufficient knowledge in this area. Examination involving different salts with cations similar to the ones specified above may be set but candidates are expected to draw out their conclusions from the observations.

N.B. No note books, course books, information booklets and text books will be allowed in the practical examination.

A learner shall be expected to perform simple calculations as outlined by the chemistry syllabus. However non programmable calculators are allowed.

PRACTICAL TECHNIQUES

Schools and students are reminded of the importance of accuracy in quantitative and qualitative exercises during the practical lessons.

- (i) A learner is expected to read the burette accurately and to the nearest volume of 0.1 cm^3 . At least 3 titrations should be done by a student to ensure a correct result and marks. Only values that fall within ± 0.2 with respect to the supervisor's volume will score full marks.
- (ii) A student is expected to take note of the temperature readings to the nearest 0.5°C . Recommended thermometer range is -10°C to 110°C . The time should be recorded in seconds and the stop clock/stop watch will be the most convenient timing instrument.
- (iii) Learner must show the ability to ignore certain values on the titration table and use only those that are consistent and compute the average of the consistent values. Consistent values must fall within 0.2 to one another.

In case of qualitative exercises a learner should use around 1 cm depth of a solution i.e. (about 2 cm^3) in a test tube. Reagents should be added drop by drop and thoroughly mixing them, to ensure effective results for each test. The student should make sure that no further changes may occur if more reagents are added. A learner should take note of the stage at which the precipitate forms and also the colour changes. Furthermore the learner must take note of chemicals used to detect gases, if any, during the experiments. Observations must be recorded as stipulated in the qualitative notes. Equations are not required during practical.

APPARATUS

The following apparatus should be stocked for teaching and examination purposes. Each learner should be provided with the necessary apparatus to conduct the experiments.

Bunsen burner

Test-tubes

Measuring cylinder calibrated 25cm^3 or 50cm^3 .

Filter funnel.

Beaker (polystyrene, glass) volume of 250cm^3 .

Conical flasks with volume of 250cm^3 .

Burette with a volume of 50cm^3 .

Pipettes with volumes of 25cm^3 or 20cm^3

Pipette fillers.

Thermometers calibrated -10°C to 110°C at intervals of 1°C .

Stop clocks/stop watches which record time in seconds.

Wash bottles.

Pyrex test tubes are essential for heating purposes with capacities $125\text{mm} \times 16\text{mm}$.

Boiling tubes i.e. of dimension $150\text{mm} \times 25\text{mm}$.

Stirring rods for stirring or mixing purposes.

Electronic balances /triple beam balances.

REAGENTS

The following standard reagents should be stocked among others. These are of paramount importance during practical.

Hydrochloric acid 1.0 mol/dm^3

Nitric acid 1.0 mol/dm^3

Sulphuric acid 0.5 mol/dm^3

Aqueous ammonia 1.0 mol/dm^3

Aqueous sodium hydroxide 1.0 mol/dm^3

Lime water (a solution of calcium hydroxide)

Aqueous silver nitrate 0.05 mol/dm^3

Aqueous potassium dichromate (VI) 0.1 mol/dm^3

Aqueous potassium iodide 0.1 mol/dm^3

Aqueous lead (II) nitrate 0.2 mol/dm^3

Aqueous potassium permanganate (VII) approximate 0.02 mol/dm^3

Barium nitrate 0.2 mol/dm^3

In addition chemical substances such as aluminium foil, red litmus paper, blue litmus paper and universal indicators should be in stock.

QUALITATIVE ANALYSIS TESTS

Notes for use in qualitative analysis

Test for anions

Anions	Test	Test result
Carbonate (CO_3^{2-})	Add dilute acid	Effervescence occurs, carbon dioxide produced
Chloride (Cl^-) [in solution]	Acidify with dilute nitric acid , then add aqueous silver nitrate	White ppt.
Iodide (I^-)[in solution]	Acidify with dilute nitric acid , then add aqueous lead (II) nitrate	Yellow ppt.
Nitrate (NO_3^-)[in solution]	Add aqueous sodium hydroxide, then aluminum foil, warm carefully.	Ammonia produced
Sulphate (SO_4^{2-}) [in solution]	Acidify with dilute nitric acid, then add aqueous barium nitrate	White ppt.

Test for aqueous cations

Cations	Effect of aqueous sodium hydroxide	Effect of aqueous ammonia
Aluminium ions (Al^{3+})	White ppt. soluble in excess giving a colourless solution	White ppt., insoluble in excess
Ammonium ions (NH_4^+)	Ammonia produced on warming	-
Calcium ions (Ca^{2+})	White ppt., insoluble in excess	No change
Copper ions (Cu^{2+})	Light blue ppt., insoluble in excess	Light blue ppt., soluble in excess, giving a dark blue solution
Iron(II) ions (Fe^{2+})	Green ppt., insoluble in excess	Green ppt., insoluble in excess, turns reddish-brown on standing
Iron (III) ions (Fe^{3+})	Red-brown ppt., insoluble in excess	Red-brown ppt., insoluble in excess
Zinc ions (Zn^{2+})	White ppt., soluble in excess giving a colourless solution	White ppt. soluble in excess giving a colourless solution.

Test for gases

Gas	Test	Test result
Ammonia	Introduce damp red litmus paper to the gas	Turns damp red litmus paper blue
Carbon dioxide	Bubble the gas through limewater	White precipitate formed
Chlorine (Cl_2)	Introduce damp blue litmus paper to the gas	Turns litmus paper red then bleaches it
Hydrogen (H_2)	Introduce a lighted splint into the gas	Puts out the lighted splint with a 'pop' sound
Oxygen (O_2)	Introduce a glowing splint into the gas	Glowing splint relighted
Sulphur dioxide (SO_2)	Bubble the gas through acidified potassium dichromate (VI)	Turns orange potassium dichromate green.

SCOPE AND SEQUENCE CHART

Topic	Grade 10		Grade 11		Grade 12
Introduction to Chemistry	SUB-TOPIC	Acids, Bases and Salts	SUB-TOPIC	Chemistry and Electricity	SUBTOPIC
	10.1.1 Introduction to Chemistry		11.1.1 Characteristic properties of acids and bases		12.1.1. Conductors
The Particulate nature of matter	10.2.1 Matter and the Kinetic theory	The mole concept	11.1.2 Preparation of salts	Metals	12.2.1 General properties of a metals
	10.2.2 Diffusion		11.2.3 Types of oxides		12.2.2 Reactivity and Electro Chemical Series
Experimental Techniques	10.3.1 Measuring of quantities		11.2.4 Identification of ions and gases (Qualitative analysis)		12.2.4 Corrosion
	10.3.2 Criteria of purity		11.2.1 Relative masses		12.2.5 Thermal stability of the compounds
	10.3.3 Separating mixtures		11.2.2 The mole	Non-Metals	12.3.1 General properties of non-metals.
Atoms, elements, molecules and compounds	10.4.1 Atomic structure and Periodic Table		11.2.3 Empirical and Molecular formulae		12.3.2. Hydrogen
	10.4.2 Bonding	Chemical reactions and energy changes	11.3.1 Rates of chemical reactions		12.3.3. Oxygen

Topic	Grade 10		Grade 11		Grade 12
	10.4.3 Macromolecules		11.3.2.Chemical equilibrium		12.3.4 Nitrogen
	10.4.4 Chemical formulae and equations		11.3.3 Redox reactions		12.3.5. Chlorine
			11.3.4 Energetics of reactions		12.3.6 Sulphur
		The Periodic Table	11.4.1 Group and the periodic trends	Organic Chemistry	12.3.7 Carbon and carbonates
			11.4.2 Group properties		12.4.1 Saturated and unsaturated Hydrocarbons

